Missile Defense Agency (MDA)
FY26 Small Business Innovation Research (SBIR) Broad Agency Announcement (BAA)
Release 4
Proposal Submission Instructions

INTRODUCTION

The Missile Defense Agency (MDA) mission is to develop and deploy a layered Missile Defense System to defend the United States, its deployed forces, allies, and friends from missile attacks in all phases of flight. The MDA SBIR Program is implemented, administered, and managed by the MDA SBIR/Small Business Technology Transfer (STTR) Program Management Office (PMO).

Proposers responding to an MDA topic in this solicitation must follow all general instructions provided in the Department of War (DoW) solicitation. MDA requirements in addition to, or deviating from, the DoW Solicitation are provided in the instructions below.

Proposers are encouraged to thoroughly review the DoW solicitation and register for the DSIP Listserv to remain apprised of important programmatic and contractual changes.

- The full DoW solicitation is available on DSIP at https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations. Be sure to select the tab for the appropriate solicitation cycle.
- Register for the DSIP Listserv at: https://www.dodsbirsttr.mil/submissions/login.

Specific questions pertaining to the administration of the MDA SBIR/STTR Program and these proposal instructions should be directed to the MDA SBIR/STTR PMO via email at mda.sbir-sttr@mail.mil.

Proposals must conform to the terms of the associated DoW solicitation and these MDA-specific requirements. MDA reserves the right to not consider any non-conforming proposals.

To receive an SBIR or STTR award, the awardee must qualify as a Small Business Concern (SBC) as defined by SBA regulations at 13 C.F.R. §§ 701-705. The eligibility requirements for the SBIR/STTR Programs are unique and do not correspond to those of other small business programs. For more information on eligibility requirements, see the Small Business Administration (SBA) SBIR/STTR website at https://www.sbir.gov/faq/eligibility-requirements

In accordance with the SBIR/STTR Policy Directive, the work within the SBIR/STTR contracts are to conduct feasibility-related experimental or theoretical Research/Research and Development (R/R&D) related to described agency requirements. The purpose of Phase I is to determine the scientific and technical merit and feasibility of the proposed effort. Phase Is are anticipated as six (6) month period of performance.

Classified Proposals

Classified proposals ARE NOT accepted under the MDA SBIR Program. The inclusion of classified data in an unclassified proposal MAY BE grounds for MDA to determine the proposal as non-responsive and the proposal not to be evaluated. Proposing SBCs currently working under any classified MDA contract must use the security classification guidance (SGC) provided under that contract to verify new SBIR proposals are unclassified prior to submission. Phase I efforts do not typically require the performer to generate, access, or store classified information.

Communication
All communication from the MDA SBIR/STTR PMO will originate from the mda.sbir-sttr@mail.mil email address. Please white-list this address in your email spam filters to ensure timely receipt of communications.

Proposal titles, abstracts, anticipated benefits, and keywords of proposals that are selected for negotiation will undergo an MDA Policy and Security Review. Proposal titles, abstracts, anticipated benefits, and keywords are subject to revision and/or redaction by MDA. Final approved versions of proposal titles, abstracts, anticipated benefits, and keywords may appear on DSIP and/or the SBA's SBIR/STTR award site (https://www.sbir.gov/awards).

MDA Cybersecurity Requirement Notice – CMMC Tier 1 (Foundational)
Small Business Concerns participating in the Missile Defense Agency (MDA) SBIR/STTR Program are required to comply with applicable DoW cybersecurity requirements for the protection of Federal Contract Information (FCI). In accordance with DFARS 252.204-7012, DFARS 252.204-7019, DFARS 252.204-7020, and DFARS 252.204-7021, MDA requests that all prospective SBIR/STTR awardees complete a CMMC Tier 1 (Foundational) self-assessment and enter the assessment results into the Supplier Performance Risk System (SPRS) prior to award. Completion of this requirement ensures appropriate safeguarding of information and supports the secure execution of MDA research, development, and technology efforts.

PHASE I PROPOSAL GUIDELINES
The DoW SBIR/STTR Innovation Portal (DSIP) is the official portal for DoW SBIR/STTR proposal submission. Proposers are required to submit proposals via DSIP; proposals submitted by any other means will be disregarded. Detailed instructions regarding registration and proposal submission via DSIP are provided in the DoW solicitation.

A complete Phase I proposal must contain the following volumes:

Proposal Coversheet (Volume 1)
The proposal coversheet is prepared on DSIP and in accordance with the DoW solicitation.

Technical Volume (Volume 2)
The technical volume is not to exceed 15 pages and must follow the formatting requirements provided in the DoW solicitation. Any pages submitted beyond the 15-page limit will not be evaluated. All acronyms should be spelled out the first time they are used within the Technical Volume. This will help avoid confusion when proposals are evaluated by technical reviewers.

Content of the Technical Volume
For technical volume format guidance, please refer to the "Format of Technical Volume" section within the DoW solicitation.

If including a Letter(s) of Support and/or Technical and Business Assistance (TABA) request, it must be included as part of Volume 5 and will not count towards the 15-page Technical Volume (Volume 2) limit. Any technical data/information that should be in the Technical Volume (Volume 2) but is contained in other Volumes will not be considered.

Cost Volume (Volume 3)
The online DSIP cost volume must be utilized. The Phase I Base amount must not exceed $307,500 (or not to exceed $314,000 if Technical and Business Assistance is included). MDA does not utilize the Phase I Option.

Please review the updated Percentage of Work (POW) calculation details included in the DoW solicitation. Proposals shall adhere to the POW calculation details included in the DoW solicitation. Occasionally, deviations from these SBIR requirements may occur, and must be approved in writing by the Contracting Officer after consultation with the MDA SBIR/STTR PMO.

Company Commercialization Report (CCR) (Volume 4)
Completion of the CCR as Volume 4 of the proposal submission in DSIP is required. Please refer to the DoW solicitation for full details on this requirement. Information contained in the CCR will not be considered by MDA during proposal evaluations.

Supporting Documents (Volume 5)
Volume 5 is provided for proposing SBCs to submit additional documentation to support the Coversheet (Volume 1), Technical Volume (Volume 2), and the Cost Volume (Volume 3). Please refer to the DoW solicitation for more information.

a. **Required:** For MDA SBIR topics that are restricted under export control regulations including the International Traffic in Arms Regulations (ITAR) and the Export Administration Regulations (EAR) offerors must provide a copy of certified DD Form 2345 or evidence of application submission. The form, instructions, and FAQs may be found at the United States/Canada Joint Certification Program website at https://www.dla.mil/Logistics-Operations/Services/JCP/DD23%2045Instructions/. Upload under Volume 5 – "Other" category.

b. ***Optional:** Letter(s) of Support. Upload under Volume 5 – "Letter of Support" category.

c. ****Optional:** Request for Technical and Business Assistance (TABA) using the MDA SBIR/STTR Phase I TABA Form (https://www.mda.mil/global/documents/pdf/SBIR_STTR_PHI_TABA_Form.pdf). Upload under Volume 5 – "Other" category.

***Letter(s) of Support:** If including letter(s) of support, they MUST be uploaded using the "Letter of Support" category within Volume 5. A qualified letter of support is from a relevant commercial or Government Agency procuring organization working with MDA, articulating their pull for the technology (i.e., what Missile Defense System need(s) the technology and why it is important to fund it), and possible commitment to provide additional funding and/or insert the technology in their acquisition/sustainment program. A letter of support shall not be contingent upon award of a subcontract. **Letters of support from any MDA officials or references to such letters in a proposal WILL NOT be accepted and may result in the rejection of the proposal.**

****Discretionary Technical and Business Assistance (TABA):** The SBIR/STTR Policy Directive allows agencies to enter into agreements with suppliers to provide technical assistance to SBIR and STTR awardees, which may include access to a network of scientists and engineers engaged in a wide range of technologies or access to technical and business literature available through on-line databases.

The purpose of this technical assistance is to assist SBIR awardees in:
1. Making better technical decisions on SBIR projects;
2. Solving technical problems that arise during SBIR projects;
3. Minimizing technical risks associated with SBIR projects;

4. Developing and commercializing new commercial products and processes resulting from such projects including intellectual property protections;
5. Providing and/or addressing cybersecurity needs; and
6. Screening for foreign involvement or influence.

A proposing SBC may acquire the technical assistance services described above on its own. Proposing SBC's must request this authority from MDA and demonstrate in its proposal that the individual or entity selected can provide the specific technical services needed. In addition, costs must be included in the Cost Volume of the proposal. The TABA provider may not be the requesting firm, an affiliate of the requesting firm, an investor of the requesting firm, or a subcontractor or consultant of the requesting firm otherwise required as part of the paid portion of the research effort (e.g. research partner or research institution).

If the proposing SBC supports the need for this requirement sufficiently as determined by the Government, MDA will permit the awardee to acquire such technical assistance, in an amount up to $6,500 per year. This will be an allowable cost on the SBIR award. The per-year amount will be in addition to the award and is not subject to any burden, profit or fee by the proposing SBC. The per-year amount is based on the original period of performance (6 months) and does not apply to period of performance extensions. Requests for TABA funding outside of the original period of performance (6 months) will not be considered.

The MDA Phase I TABA form can be accessed here: (https://www.mda.mil/global/documents/pdf/SBIR_STTR_PHI_TABA_Form.pdf) and must be included as part of Volume 5 using the "Other" category.

All requests for TABA must be completed using the MDA SBIR/STTR Phase I TABA Form and included as a part of Volume 5. MDA will not accept requests for TABA that do not utilize the MDA SBIR/STTR Phase I TABA Form or are not provided as part of Volume 5.

Fraud, Waste and Abuse Training (Volume 6)
Fraud, Waste and Abuse training material can be found in the Volume 6 section of the proposal submission module in DSIP and must be thoroughly reviewed once per year to proceed with proposal submission.

Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)
Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. Please be aware that the Disclosures of Foreign Affiliations or Relationships to Foreign Countries WILL NOT be accepted as a PDF Supporting Document in Volume 5 of the DSIP proposal submission. Do not upload any previous versions of this form to Volume 5. For additional details, please refer to the DoW solicitation.

PHASE II PROPOSAL GUIDELINES
Phase II proposals may only be submitted by Phase I awardees. Details on the due date, format, content, and submission requirements of the Phase II proposal will be provided to Phase I awardees by the MDA SBIR/STTR PMO during the fourth (4^{th}) month of the Phase I period of performance.

MDA will evaluate and select Phase II proposals using the Phase II evaluation criteria listed in the DoW solicitation. While funding must be based upon the results of work performed under a Phase I award and the scientific and technical merit, feasibility and commercial potential of the Phase II proposal, Phase I final reports will not be reviewed as part of the Phase II evaluation process. The Phase II proposal should include a concise summary of the Phase I effort including the specific technical problem or opportunity addressed and its importance, the objective of the Phase I effort, the type of research conducted, findings or results of this research, and technical feasibility of the proposed technology. Due to funding limitations, MDA reserves the right to limit awards under any topic and only those proposals of superior scientific and technical quality, as determined by MDA, will be funded.

Phase II efforts may be awarded as firm-fixed-price or cost-reimbursement type contracts. Phase II awardees proposing a cost-reimbursement type contract must have an accounting system determined adequate by the Defense Contract Management Agency (DCMA). It is strongly encouraged that an adequate accounting system be in place prior to the Phase II proposal submission timeframe. Lack of an adequate accounting system will delay and/or prevent Phase II contract award. For more information on Defense Contract Audit Agency (DCAA) accounting system audits and requirements visit https://www.dcaa.mil/Customers/Small-Business.

Classified proposals are not accepted under the DoW solicitation. If topics will require classified work during Phase II, the proposing SBC must have the appropriate facility clearance (FCL). For more information on facility and personnel clearance procedures and requirements, visit the Defense Counterintelligence and Security Agency (DCSA) website at: https://www.dcsa.mil/mc/ctp/fc/.

EVALUATION AND SELECTION

All proposals will be evaluated in accordance with the evaluation criteria listed in the DoW solicitation and the guidance provided in these instructions. Selections will be based on best value to the Government considering the evaluation criteria listed in the DoW solicitation which are listed in descending order of importance.

Technical reviewers will base their conclusions only on information contained in the proposal. It should not be assumed that reviewers are acquainted with the firm or key individuals or any referenced experiments. Relevant supporting data such as journal articles, literature, including Government publications, etc., should be listed in the proposal and will count toward the applicable page limit.

MDA reserves the right to select none, one, or more than one proposal under any topic. MDA is not responsible for any money expended by the proposing SBC in responding to this solicitation. Due to funding limitations, MDA reserves the right to limit awards under any topic and only those proposals of superior scientific and technical quality, as determined by MDA, will be funded. MDA reserves the right to withdraw from negotiations at any time prior to award for any reason including matters of national security (foreign persons, foreign influence or ownership, inability to clear the firm or personnel for security clearances, or other related issues).

AWARD AND CONTRACT INFORMATION

Contracting Officer

The MDA Contracting Office will issue any resulting contract awards. The MDA Contracting Officer is the only Government official authorized to enter into any binding agreement or contract on behalf of the Government.

Contract Type

It is anticipated that all Phase I efforts will be awarded as a firm-fixed-price contract. The period of performance is six (6) months to include submission and Government acceptance of all final deliverables.

Deliverables

A Phase I effort is not intended to have any formal end-item contract delivery nor ownership by the Government of the awardee’s hardware, computer software, or rights in technical data. As a result, proposals should not contain any reference to the term "Deliverables" when referring to hardware, computer software, or technical data. Instead use the term: “Products for Government Testing, Evaluation, Demonstration, and/or possible destructive testing.”

The standard deliverables for a Phase I are:
- Report of Invention(s), Contractor, and/or Subcontractor(s) // Patent Application for Invention
- Status Report // Bi-monthly Status Report
- Contract Summary Report // Final Report
- Certification of Compliance // Life Cycle Certification
- Computer Software Product // Product Description (if applicable)

ADDITIONAL INFORMATION

Notices and Feedback

Firms will be notified of selection or non-selection for negotiation towards a Phase I award within ninety (90) days of the closing date of the solicitation. The MDA SBIR/STTR PMO will distribute selection and non-selection email notices to all firms who submit an MDA SBIR proposal. Notification of selection or non-selection will be provided via email from the mda.sbir-sttr@mail.mil email address and sent to the Corporate Official and Principal Investigator identified on the Phase I proposal. MDA is not responsible for firms that provide incorrect contact information or changes to such information after proposal submission.

MDA will provide written feedback to unsuccessful offerors regarding their proposals upon request. Requests for feedback must be submitted in writing to the MDA SBIR/STTR PMO within 30 calendar days of non-selection notification. Non-selection notifications will provide instructions for requesting proposal feedback. Only firms that receive a non-selection notification are eligible for written feedback.

Refer to the DoW solicitation for procedures on pre-award and post-award protests. For the proposes of a protest related to a particular topic selection, non selection, or award decisions, protests should be served to the MDA SBIR/STTR Program Manager, Candace Wright at mda.sbir-sttr@mail.mil.

Performance Benchmark Requirements for Phase I Eligibility

MDA does not accept proposals from firms that are currently ineligible for Phase I awards as a result of failing to meet the benchmark rates at the last assessment. Additional information on Benchmark Requirements can be found in the DoW solicitation.

Use of Federally Funded Research and Development (FFRDC) and Support Contractors

Only Government personnel with active non-disclosure agreements (NDA) will evaluate proposals. Non-Government technical consultants (“consultants”) to the Government may review and provide support during proposal evaluations. Non-Government consultants may have access to the proposals, may be utilized to review proposals, and may provide comments and recommendations to the Government decision makers. Non-Government consultants will not make decisions regarding the selection or non-selection of a proposal. They are also expressly prohibited from competing for MDA SBIR awards in the SBIR topics they review and/or on which they provide advisement to the Government.

All Non-Government consultants are required to comply with procurement integrity laws. These consultants will not have access to proposals or pages of proposals that are properly labeled by the offeror as "Government Only." Pursuant to FAR 9.505-4, the MDA contracts with the organizations to which these consultants are employed, include a clause which requires them to (1) protect the offerors’ information from unauthorized use or disclosure for as long as it remains proprietary and (2) refrain from using the information for any purpose other than that for which it was furnished. In addition, MDA requires all Consultants to execute NDAs. These NDAs remain on file with the MDA SBIR/STTR PMO.

Non-Government consultants will only be authorized to access those portions of the proposal data and participate in discussions that are necessary to enable them to perform their respective duties. In accomplishing their duties related to the evaluation process, consultants may require access to proprietary information contained in the offerors' proposals.

Organizational Conflict of Interest (OCI)
The general OCI rules for contractors that support development and oversight of SBIR topics are covered in FAR 9.505-1 through FAR 9.505-4 and DFARS 209.505 through DFARS 209.571-8 as the means of avoiding, neutralizing, or mitigating organizational conflicts of interest. All applicable rules under the FAR 9.5 and DFARS 209.5 apply.

If you, or another employee in your company, developed or assisted in the development of any SBIR requirement or topic, please be advised that your company may have an OCI. Your company could be precluded from an award under this solicitation if your proposal contains anything directly relating to the development of the requirement or topic. Before submitting your proposal, please examine any potential OCI issues that may exist with your company to include subcontractors. Understand that if any actual or potential OCI exist, your company may be required to submit an acceptable OCI mitigation plan prior to award.

In addition, FAR 3.101-1 states that Government business shall be conducted in a manner above reproach and, except as authorized by statute or regulation, with complete impartiality and with preferential treatment for none. The general rule is to avoid strictly any conflict of interest or even the appearance of a conflict of interest in Government-contractor relationships. An appearance of impropriety may arise where an offeror may have gained an unfair competitive advantage through its hiring of, or association with, a former Government official if there are facts indicating the former Government official, through their former Government employment, had access to non-public, competitively useful information. (See *Health Net Fed. Svcs*, B-401652.3; *Obsidian Solutions Group*, LLC, B-417134, 417134.2). The existence or the appearance of an unfair competitive advantage may result in an offeror being disqualified and this restriction cannot be waived.

It is MDA policy to ensure all appropriate measures are taken to resolve OCIs arising under FAR 9.5 and unfair competitive advantages arising under FAR 3.101-1 to prevent the existence of conflicting roles that might bias a contractor’s judgment and deprive MDA of objective advice or assistance, and to prevent contractors from gaining an unfair competitive advantage.

Foreign Nationals
Through submission of a proposal, the proposing SBC acknowledges access to MDA Controlled Unclassified Information (CUI) and legacy For Official Use Only (FOUO) shall be limited to U.S. Persons and dual citizens that do not violate the Countries list in 22 CFR 126.1. Foreign Nationals (as defined in the DoW solicitation) will not be approved for access to MDA CUI and legacy FOUO. Exceptions may be granted in limited circumstances by the Contracting Officer in coordination with the MDA SBIR/STTR PMO.

Export Control Regulations
All MDA SBIR topics are restricted under export control regulations including the International Traffic in Arms Regulations (ITAR) and the Export Administration Regulations (EAR). ITAR controls the export and import of listed defense-related material, technical data and services that provide the United States with a critical military advantage. EAR controls military, dual-use and commercial items not listed on the United States Munitions List or any other export control lists. EAR regulates export controlled items based on user, country, and purpose. The proposing SBC must ensure their firm complies with all

applicable export control regulations. Refer to the following URLs for additional information: https://www.pmddtc.state.gov/ and https://www.bis.doc.gov/index.php/regulations/export-administration-regulations-ear.

Data Assertions
10 U.S.C. 3772(a) requires, to the maximum extent practicable, an identification prior to delivery of any technical data (and computer software) to be delivered to the Government with restrictions on use. Should a proposal be selected for award, the proposing SBC and its subcontractors will be responsible to identify any technical data and/or computer software that will be delivered to the Government under the effort with restrictions on use. The offeror's rights in technical data and/or computer software delivered, developed or generated during performance of this SBIR effort are detailed in DFARS 252.227-7018 – current version can be found here: https://www.acquisition.gov/dfars/part-252-solicitation-provisions-and-contract-clauses

FLOW-DOWN CLAUSES

The following MDA Local contract clauses apply to the prime offeror and must be flowed to any research partners (including universities and educational institutions) and/or subcontractors on the effort. Note: This list does not include all contract clauses which must be flowed to the subcontract level. The prime offeror is responsible for the review of all contract terms and conditions.

MDA H-08 PUBLIC RELEASE OF INFORMATION (AUG 2024)

a. In addition to the requirements of 32 CFR § 117, NISPOM Rule, any official MDA information/materials that a contractor or subcontractor intends to release to the public that pertains to any work under performance of this contract must receive a pre-publication review by the Missile Defense Agency (MDA) and must be authorized for release by MDA. At a minimum, these information/materials may be technical papers, presentations, articles for publication, key messages, talking points, speeches, and social media or digital media, such as press releases, photographs, fact sheets, advertising, posters, videos, etc.

b. Subcontractor public information/materials must be submitted for approval through the prime contractor to MDA.

c. Upon request to the MDA Procuring Contracting Officer (PCO), contractors shall be provided the "Request for Industry Media Engagement" form (or any superseding MDA form).

d. At least 45 calendar days prior to the desired release date, the contractor must submit the required form and information/materials to be reviewed for public release to the appropriate MDA Program Office and simultaneously provide courtesy copy to the appropriate PCO and MDAPressOperations@mda.mil. (Additional distribution emails can be added by the Program Office to ensure proper internal coordination and tracking of public release requests.)

e. All information/materials submitted for MDA review must be an exact copy of the intended item(s) to be released, must be of high quality, and free of tracked changes and/or comments. Photographs must have captions, and videos must have the intended narration included. All items must be marked with the applicable month, day, and year.

f. No documents or media shall be publicly released by the Contractor without MDA Public Release approval.

g. Once information has been cleared for public release, it resides in the public domain and must always be used in its originally cleared context and format. Information previously cleared for public release but containing new, modified or further developed information must be re-submitted in accordance with this clause.

(End of Clause)

MDA H-09 ORGANIZATIONAL CONFLICT OF INTEREST (APR 2020)

a. Purpose: The purpose of this clause is to ensure that:

(1) the Contractor is rendering impartial assistance and advice to the Government at all times under this contract and related Government contracts;

(2) the Contractor’s objectivity in performing work under this contract or related Government contracts is not impaired; and

(3) the Contractor does not obtain an unfair competitive advantage by virtue of its access to non-public Government information, or by virtue of its access to proprietary information belonging to others.

b. Scope: The Organizational Conflict of Interest (OCI) rules, procedures and responsibilities described in FAR 9.5 “Organizational and Consultant Conflicts of Interest”, FAR 3.101-1 “Standards of Conduct – General, DFARS 209.5 “Organizational and Consultant Conflicts of Interest,” and in this clause are applicable to the prime Contractor (including any affiliates and successors-in-interest), as well as any co-sponsor, joint-venture partner, consultant, subcontractor or other entity participating in the performance of this contract. The Contractor shall flow this clause down to all subcontracts, consulting agreements, teaming agreements, or other such arrangements which have OCI concerns, while modifying the terms "contract", "Contractor", and "Contracting Officer" as appropriate to preserve the Government's rights.

c. Access to and Use of Nonpublic Information: If in performance of this contract the contractor obtains access to nonpublic information such as plans, policies, reports, studies, financial plans, or data which has not been released or otherwise made available to the public, the Contractor agrees it shall not use such information for any private purpose or release such information without prior written approval from the Contracting Officer.

d. Access to and Protection of Proprietary Information: The Contractor agrees to exercise due diligence to protect proprietary information from misuse or unauthorized disclosure in accordance with FAR 9.505-4. The Contractor may be requested to enter into a written non-disclosure agreement with a third party asserting proprietary restrictions, if required in the performance of the contract.

e. In accordance with FAR 3.101-1, the Contractor shall also take all appropriate measures to prevent the existence of conflicting roles that might bias the Contractor’s judgement, give the Contractor an unfair competitive advantage, and deprive MDA of objective advice or assistance that can result from hiring former Government employees. (See Health Net Fed. Svcs, B-401652.3).

f. Restrictions on Participating in Other Government Contract Efforts. <<<Contracting Officer fill-in. If there are no restrictions, then fill in “NONE”.>>>

g. OCI Disclosures: The Contractor shall disclose to the Contracting Officer all facts relevant to the existence of an actual or potential OCI, using an OCI Analysis/Disclosure Form which the Contracting

Officer will provide upon request. This disclosure shall include a description of the action the Contractor has taken or plans to take to avoid, neutralize or mitigate the OCI.

h. Remedies and Waiver:

(1) If the contractor fails to comply with any requirements of FAR 9.5, FAR 3.101-1, DFARS 209.5, or this clause, the Government may terminate this contract for default, disqualify the Contractor from subsequent related contractual efforts if necessary to neutralize a resulting organizational conflict of interest, and/or pursue other remedies permitted by law or this contract. If the Contractor discovers and promptly reports an actual or potential OCI subsequent to contract award, the Contracting Officer may terminate this contract for convenience if such termination is deemed to be in the best interest of the Government, or take other appropriate actions.

(2) The parties recognize that the requirements of this clause may continue to impact the contractor after contract performance is completed, and that it is impossible to foresee all future impacts. Accordingly, the Contractor may at any time seek an OCI waiver from the Director, MDA by submitting a written waiver request to the Contracting Officer. Any such request shall include a full description of the OCI and detailed rationale for the OCI waiver.

(End of Clause)

MDA H-28 DISTRIBUTION CONTROL OF TECHNICAL INFORMATION (AUG 2014)

a. The following terms applicable to this clause are defined as follows:

1. DoW Official. Serves in DoW in one of the following positions: Program Director, Deputy Program Director, Program Manager, Deputy Program Manager, Procuring Contracting Officer, Administrative Contracting Officer, or Contracting Officer's Representative.

2. Technical Document. Any recorded information (including software) that conveys scientific and technical information or technical data.

3. Scientific and Technical Information. Communicable knowledge or information resulting from or pertaining to the conduct or management of effort under this contract. (Includes programmatic information).

4. Technical Data. As defined in DFARS 252.227-7013.

b. Except as otherwise set forth in the Contract Data Requirements List (CDRL), DD Form 1423 the distribution of any technical documents prepared under this contract, in any stage of development or completion, is prohibited outside of the contractor and applicable subcontractors under this contract unless authorized by the Contracting Officer in writing. However, distribution of technical data is permissible to DOW officials having a "need to know" in connection with this contract or any other MDA contract provided that the technical data is properly marked according to the terms and conditions of this contract. When there is any doubt as to "need to know" for purposes of this paragraph, the Contracting Officer or the Contracting Officer's Representative will provide direction. Authorization to distribute technical data by the Contracting Officer or the Contracting Officer's Representative does not constitute a warranty of the technical data as it pertains to its accuracy, completeness, or adequacy. The contactor shall distribute this technical data relying on its own corporate best practices and the terms and conditions of this contract. Consequently, the Government assumes no responsibility for the distribution of such technical data nor will the Government have any liability, including third party liability, for such technical data should it be inaccurate, incomplete, improperly marked or otherwise defective. Therefore, such a distribution shall not violate 18 United States Code § 1905.

c. All technical documents prepared under this contract shall be marked with the following distribution statement, warning, and destruction notice identified in sub-paragraphs 1, 2 and 3 below. When it is technically not feasible to use the entire WARNING statement, an abbreviated marking may be used, and a copy of the full statement added to the "Notice To Accompany Release of Export Controlled Data" required by DoW Directive 5230.25.

1. DISTRIBUTION - [PCO, Insert the appropriate distribution statement and complete the statement, if necessary, to include the applicable controlling office.]

2. WARNING - This document contains technical data whose export is restricted by the Arms Export Control Act (Title 22, U.S.C., Sec 2751, et seq.) or the Export Administration Act of 1979 (Title 50, U.S.C., App. 2401 et seq), as amended. Violations of these export laws are subject to severe criminal penalties. Disseminate in accordance with provisions of DoW Directive 5230.25

3. DESTRUCTION NOTICE - For classified documents follow the procedures in DOW 5220.22-M, National Industrial Security Program Operating Manual, February 2006, Incorporating Change 1, March 28, 2013, Chapter 5, Section 7, or DoWM 5200.01-Volume 3, DoW Information Security Program: Protection of Classified Information, Enclosure 3, Section 17. For controlled unclassified information follow the procedures in DoWM 5200.01-Volume 4, Information Security Program: Controlled Unclassified Information.

d. The Contractor shall insert the substance of this clause, including this paragraph, in all subcontracts.

(End of Clause)

MDA SBIR DOW 2026 BAA
Topic Index
Release 4

MDA26BZ04-NV001 TITLE: Next-Gen Thermal Batteries for Missile Defense Application

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Contested Logistics Technologies (LOG); Scaled Hypersonics (SHY); Scaled Directed Energy (SCADE)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Hypersonics; Materials; Battlespace

PROJECTED CMMC LEVEL REQUIREMENT: Level 1

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop and demonstrate advanced thermal battery technology that significantly reduces size and weight while extending operational life, enabling enhanced performance and mission capabilities for missile defense systems.

DESCRIPTION: Current thermal battery technologies, while highly reliable for critical power applications in missile defense, present limitations that directly impact the performance and capabilities of advanced interceptor systems. Key constraints include their specific energy density, power density, size, weight, and operational lifespan. These factors limit the overall effectiveness and maneuverability of these crucial defense assets. For example, the Standard Missile 3 (SM-3) currently utilizes multiple discrete Lithium thermal batteries to power its various subsystems, each with specific voltage, current, and duration requirements. The size, weight, and performance of these batteries are dictated by the manufacturer (Enersys, EaglePicher, and GYT) and the demands of the design, but there is a clear need for improved performance. A single, more efficient battery solution could greatly simplify the power architecture, reduce weight, and enhance system responsiveness.

This SBIR topic seeks innovative research and development to create next-generation thermal batteries that overcome these limitations. We are looking for proposals focused on achieving significant improvements in specific energy density, power density, operational lifespan, and miniaturization. Ideally, solutions would be lightweight, exceptionally safe, and designed to operate reliably in the harsh environments typical of missile defense deployments. Innovations may include, but are not limited to, novel electrolyte chemistries, advanced electrode materials, improved thermal management techniques, and alternative activation methods. The ultimate goal is a new generation of thermal batteries that enable enhanced missile defense capabilities through improved performance, reduced system footprint, and extended operational readiness.

PHASE I: Phase I would focus on conducting a trade study to identify the most promising candidate materials and designs for smaller dimension, lightweight, and extended-duration thermal batteries. This would include a feasibility analysis, a cost estimate, and the identification of candidate solution sets that use measurable metrics (e.g., specific energy, specific power, volume, weight, lifespan) to provide equivalent or better solutions than existing technologies.
Proposed Deliverable:
-Trade study report

-Modeling and simulation results
-Prototype cell test data
-Thermal management model
-Manufacturing feasibility assessment
-Detailed Phase II plan

PHASE II: Phase II plan is to construct a prototype of the best candidate thermal battery design from Phase I and test it to qualify it as a potential replacement for current thermal batteries used in missile defense systems. This would involve optimizing the battery design, improving manufacturing processes, and conducting comprehensive performance and safety testing. The prototype would be tested under simulated operational conditions to validate its performance and reliability.

PHASE III DUAL USE APPLICATIONS: The technologies developed during this SBIR, such as novel electrolytes, advanced electrode materials, and improved thermal management, could be translated to several civilian and other defense applications, creating a substantial dual-use market.

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1. J. Edington,Z. Van Zandt, k. Haines, I. Mills, Novel Materials for High Voltage Thermal Batteries, CFD Research Corp and ARMDEC, US Army RDECOM: https://www.powersourcesconference.com/Power%20Sources%202018%20Digest/docs/25-1.pdf
2. Sandia National Labs, New Thermal Battery Manufacturing Method: https://newsreleases.sandia.gov/thermal-battery/
3. S. Roberts , Development of cloud-based tool for predicting thermal battery performance: TABS v6: https://www.powersourcesconference.com/PowerSources23/docs/13-2.pdf

KEYWORDS: Thermal Batteries; Missile Defense; smaller; lightweight; higher density; longer duration; enhance performance

MDA26BZ04-NV002 TITLE: Radiation Hardening of Non-Hardened Commercial Microelectronics

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Quantum and Battlefield Information Dominance (Q-BID)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Microelectronics; Space Technology

PROJECTED CMMC LEVEL REQUIREMENT: Level 1

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OBJECTIVE: Develop a process to radiation harden commercial microelectronics that were not originally designed to operate in radiation environments.

DESCRIPTION: The performance and survivability of Department of War (DoW) and space systems, especially those operating in space environments, are critically affected by radiation. Radiation testing of parts is a major cost and schedule driver for DoW and space systems. The market for microelectronics that meet government radiation requirements is small. Commercial microelectronics could meet the government's performance requirements while failing to meet its natural space requirements. Some state-of-the-art processes like Gate All Around (GAA) technologies meet the DoW's performance and Size, Weight, and Power (SWAP) requirements yet might not meet all of the DoW's survivability requirements. MDA seeks a process which utilizes advanced packaging techniques and chiplets to pair fast 'non-hardened' components with slower hardened packages together, thereby increasing the radhard (radiation hardened) tolerance while maintaining the high performance of non-hardened commercial technologies. The goal is to target a minimum Single Event Latch-up (SEL) immunity of 75 Linear Energy Transfer (LET) and Total Ionizing Dose (TID) survival of 300 kRad(Si).

This topic will seek to take existing Commercial Off-The-Shelf (COTS) parts/chiplets fabricated using state-of-the-art processes such as sub-16nm FinFETs (Fin Field Effect Transistor) or GAA modify them or provide a package on package like solution such that their radiation tolerance meets natural space requirements. This includes solutions based on heterogeneous packaging, where a radhard "watchdog" chiplet is integrated to monitor and compensate for radiation-induced errors in the high-performance COTS component. At a minimum, the final product should meet an SEL immunity requirement of at least 75 LET and must be able to survive at least a 300 kRad TID. Solutions relying on shielding the part or requiring access at the state-of-the-art foundry to add or change masks are not of interest. All solutions should start with already fabricated parts either in bare die form or packaged parts.

PHASE I: Feasibility Study and Proof of Concept: Develop the proposed approach to a sufficient level to demonstrate its viability and identify requirements for full development. This should include detailed simulation and modeling of the proposed heterogeneous packaging architecture, showing predicted radiation tolerance improvements and potential performance impacts. Explore different integration schemes for the hardened and non-hardened chiplets.

Component Selection and Characterization: Identify suitable COTS components (e.g., FinFETs, Gate All Around (GAA), Fully Depleted Silcon on Insulator (FD-SOI) devices) and chiplets for the heterogeneous packaging approach. Perform baseline radiation testing on the selected COTS components to quantify their initial radiation tolerance.
Design and Simulation: Design the initial heterogeneous package, including interconnects, thermal management, and power distribution. Simulate the radiation response of the packaged system, considering both TID and Single-Event Upset (SEU) effects.
Deliverables:
-Detailed feasibility study report outlining the proposed approach, simulation results, and component selection rationale.
-A preliminary design of the heterogeneous package.
-A test plan for Phase II radiation testing.

PHASE II: -Prototype Development and Fabrication: Fabricate a prototype heterogeneous package based on the Phase I design. This may involve collaboration with a packaging vendor or trusted foundry.
-Radiation Testing and Optimization: Conduct comprehensive radiation testing (TID, SEU, SEL, and dose rate) of the fabricated prototype to characterize its radiation tolerance. Optimize the design and fabrication process based on the test results.
-Performance Evaluation: Evaluate the performance of the heterogeneous package in terms of speed, power consumption, and signal integrity. Compare the performance to that of the original COTS component.
-Integration and Validation: Integrate the hardened component into a representative space avionic subsystem/system application. Test in realistic space radiation environments.

PHASE III DUAL USE APPLICATIONS: Deliverables:
-Fabricated and tested prototype heterogeneous package.
-Detailed radiation test report.
-Performance evaluation report.

REFERENCES:

1. J. R. Srour and J. M. McGarrity, "Radiation effects on microelectronics in space," in Proceedings of the IEEE, vol. 76, no. 11, pp. 1443-1469, Nov. 1988, doi: 10.1109/5.90114.
2. A. H. Johnston, "Radiation effects in advanced microelectronics technologies," in IEEE Transactions on Nuclear Science, vol. 45, no. 3, pp. 1339-1354, June 1998, doi: 10.1109/23.685206.
3. P. Nsengiyumva et al., "Analysis of Bulk FinFET Structural Effects on Single-Event Cross Sections," in IEEE Transactions on Nuclear Science, vol. 64, no. 1, pp. 441-448, Jan. 2017, doi: 10.1109/TNS.2016.2620940.
4. R. Liu et al., "Single Event Transient and TID Study in 28 nm UTBB FDSOI Technology," in IEEE Transactions on Nuclear Science, vol. 64, no. 1, pp. 113-118, Jan. 2017, doi: 10.1109/TNS.2016.2627015.

KEYWORDS: Radiation; Space Weather; Sensors; Electronics; Space Platforms; Single Event Effects; Total Ionizing Dose; Space Systems; Heavy Ion; Proton; Chiplets; Gate All Around; FinFET; Advanced Packaging; Interconnect

MDA26BZ04-NV003 TITLE: Software Defined Antenna for Interceptor-to-Interceptor Communications

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Quantum and Battlefield Information Dominance (Q-BID)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Hypersonics; Microelectronics; Advanced Materials

PROJECTED CMMC LEVEL REQUIREMENT: Level 1

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OBJECTIVE: Develop low-cost software defined circular-band (ring shape) antenna for near-omnidirectional missile to missile communication in high-speed endo-atmospheric and exo-atmospheric environments.

DESCRIPTION: This topic seeks development of a flight antenna capable of multiple node, simultaneous, high-speed, secure communication in meshed High Assurance Internet Protocol Encryptors (HAIPE) "Zero Trust" platforms and networks. System shall provide wideband, omnidirectional-like (quasi-isotropic) coverage with beam steering and shaping flexibility with multiple beam transmit and receive capability. Antenna shall address frequency hopping, band switching, and modulation manipulation capabilities. Antenna shall have the capabilities of multiple antenna reception and transmission polarization associated with flight. Antenna shall have the ability to maintain secure communication while in contested spectrums. Antenna shall be able to operate in S+ to Ka frequency bands, and transmit 250 km in exo-atmospheric and endo-atmospheric hypersonic environments. For purposes of this topic, assume missile diameter of 10”. Solutions shall minimize the size, weight, power, and cost.

PHASE I: Design and develop innovative solutions, methods, and concepts for advanced software defined antenna capable of harsh flight survivability and operations. Present data supporting initial design through modeling, experimentation, and/or testing.

PHASE II: - Refine and document detailed requirements in collaboration with the interceptor system integrator.
- Develop a functional antenna prototype utilizing existing Software Defined Radio (SDR) assets.
- Conduct comprehensive testing of the antenna prototype in representative operational environments.
- Demonstrate successful antenna operation and data acquisition within the target system.

PHASE III DUAL USE APPLICATIONS: - Define clear communication requirements and interface specifications in collaboration with the system integrator.
- Establish a functional communication link between two representative missile platforms.
- Demonstrate successful data exchange between the interceptor, validating interoperability.
- Assess communication range, latency, and data throughput.

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1. A. Narbudowicz, M. J. Ammann, M. Plotka, L. Kulas, K. Nyka and M. Rzymowski, "Compact antenna for digital beamforming with software defined radios," 2017 International Symposium on Antennas and Propagation (ISAP), Phuket, Thailand, 2017, pp. 1-2, doi: 10.1109/ISANP.2017.8228737.
2. N. Sheremet, G. Fokin, D. Volgushev and V. Grigoriev, "Software Defined Radio Four-Element Antenna Array Design with Nanosecond Synchronisation," 2025 Systems of Signals Generating and Processing in the Field of on Board Communications, Moscow, Russian Federation, 2025, pp. 1-5, doi: 10.1109/IEEECONF64229.2025.10948085
3. J. Wang and K. Mouthaan, "Robust Beamforming for Conformal Antenna Arrays using Software Defined Radio," 2021 IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting (APS/URSI), Singapore, Singapore, 2021, pp. 1485-1486, doi: 10.1109/APS/URSI47566.2021.9704459.

KEYWORDS: loop antenna; software defined; interceptor communications; missile communications; quasi-isotropic; HAIPE; flight antenna; beam steering; hypersonic; secure communications; frequency hopping; band switching; modulation manipulation

MDA26BZ04-NV004 TITLE: Potential Application of Very Large Array (VLA) over Transmission Infrastructure

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Quantum and Battlefield Information Dominance (Q-BID)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Trusted AI and Autonomy; Integrated Sensing and Cyber; Quantum Science; Space Technology

PROJECTED CMMC LEVEL REQUIREMENT: Level 1

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OBJECTIVE: The application of Very Large Array (VLA) techniques to the utilization of existing transmission infrastructure.

DESCRIPTION: The Missile Defense Agency (MDA) seeks to add multiple resilient layers to the Missile Defense System. The proposed system uses current power infrastructure to detect potential anomalies generated by unknown sources.

PHASE I: While solutions may incorporate any combination of hardware, software, and infrastructure upgrades, proposals would be evaluated to the extent that they leverage existing infrastructure with minimal changes required. The solution’s feasibility should include innovative techniques to optimize the anticipated results. Phase I Deliverables should include:

- Assessment of the challenges to performance and optimization techniques
- A roadmap to develop a Technology Readiness Level (TRL) 6 product
- Identify new infrastructure required to achieve TRL 6, if any
- Development, Test, and Evaluation Plan of a TRL 6 Prototype

PHASE II: The overall mission demonstrates the ability to coherently receive and transmit from multiple geographically distributed infrastructures. Proposals will be evaluated to the degree that they deliver enabling technologies to the detection and tracking of unknown sources. Evaluation will also consider to what degree the suggested technology utilizes existing infrastructure and enables the realization of VLA measurements. Demonstrate real-time electric and magnetic field mapping from ground to endo-atmosphere with potential relativistic effects. MDA seeks leveraging existing infrastructure and biology inspired neuromorphic data processing along with advanced techniques to turn grids into sensing arrays. The vendor solution should culminate in a demonstration, subject to conditions to verify function and performance.

PHASE III DUAL USE APPLICATIONS: Conduct engineering and manufacturing development, test, and evaluation in a realistic environment with a system level test-bed. All technical parameters for the given technology should be verified, along with a revolutionary leap forward in sensing capabilities.

REFERENCES:

1. https://www.dvidshub.net/video/833546/crrel-tests-tripline-capabilities-during-ae22*
2. https://technology.nasa.gov/patent/GSC-TOPS-141
3. https://www.nature.com/articles/s44384-025-00007-8
4. https://www.cto.mil/wp-content/uploads/2025/03/TRA-Guide-Feb2025.v2-Cleared.pdf (Page 7) "Examples include testing a prototype in a high-fidelity laboratory environment or in a simulated operational environment" – Page 7

KEYWORDS: Distributed Nodes; Artificial Intelligence; sensor; communications

MDA26BZ04-NV005 TITLE: Miniaturization, Modulation, and Chip Scaling of Photonic Technologies

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Quantum and Battlefield Information Dominance (Q-BID)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software; Integrated Sensing and Cyber; Microelectronics; Quantum Science; Space Technology; Advanced Materials

PROJECTED CMMC LEVEL REQUIREMENT: Level 1

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OBJECTIVE: To advance and adopt chip-scale photonic and quantum technologies for operation in the Missile Defense System.

DESCRIPTION: The Missile Defense Agency seeks advancing and incorporating quantum and photonic technologies that can augment current state-of-the-art radar and communication systems into fielded hardware. Photonic technologies coupled with measurable quantum effects could dramatically increase the resilience and effectiveness of the sensors and radios in the Missile Defense System. Selected proposals shall provide a pathway to common building block technologies to direct sample and/or heterodyne structure software defined radios (SDRs). Low size, weight, and power (SWaP) photonic and quantum transceivers that are compatible with modern sensor requirements will be prioritized.

PHASE I: Phase I work should address the challenges and inhibiting factors for large-scale adaptation of photonic and quantum technologies. Operation in real-world environments, scalability to larger arrays, transmit capabilities, laser SWaP issues, compatible back-end interfaces, chip scale (miniaturization) and manufacturability should be evaluated. Optional and selectable components for a possible Phase II effort should be included. Phase I Deliverables should include:

- Assessment of the challenges to performance, capability, scalability, and packaging
- A roadmap to develop a Technology Readiness Level (TRL) 6 product
- Identify hardware, software, and material required in Phase II
- Development, Test, and Evaluation Plan of a TRL 6 Prototype

PHASE II: Successful performance shall culminate with operation in real-world environments in such a way that it addresses the scalability, transmit, and SWaP challenges evaluated in Phase I. Proposals should have a clear transition path forward with partners capable of chip-scaling novel quantum and photonic sensing architectures and integrating those technologies into the Missile Defense System. Conduct engineering and manufacturing development, test, evaluation in a realistic environment with a system level testbed.

PHASE III DUAL USE APPLICATIONS: Production, test, and evaluation in a realistic environment with a system level testbed. Technology shall be ready for demonstration in relevant operational environment. This environment could include placement on airborne or space low SWaP platforms. All technical parameters for the technology given will be verified. Production and manufacturing should be scalable with a miniaturized package.

REFERENCES:

1. https://www.whitehouse.gov/wp-content/uploads/2025/03/Amended-National-Strategy-on-Microelectronics-Research.pdf (Page 18)
   "Advanced photonics are poised to deliver dedicated artificial intelligence/machine learning (AI/ML) hardware that operates at low power and extraordinary speed."- Page 18
2. https://www.darpa.mil/research/programs/generating-rf-with-photonic-oscillators
3. https://www.militaryaerospace.com/sensors/article/14302371/raytheon-technologies-corp-photonic-radar-prototype
4. https://www.cto.mil/wp-content/uploads/2025/03/TRA-Guide-Feb2025.v2-Cleared.pdf (Page 7)
   "Examples include testing a prototype in a high-fidelity laboratory environment or in a simulated operational environment" – Page 7

KEYWORDS: Distributed; sensor; secure communications; photonics; microelectronics; software defined radio; quantum effects

MDA26BZ04-NV006 TITLE: Neuromorphic Hardware

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Biomanufacturing (BIO)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Trusted AI and Autonomy; Advanced Computing and Software; Integrated Sensing and Cyber; Microelectronics; Space Technology;

PROJECTED CMMC LEVEL REQUIREMENT: Level 1

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OBJECTIVE: Realize next-generation neuromorphic technology to enable continuous adaptation and self-optimization to maintain overmatch in Electronic Warfare (EW) and cyber-contested domains.

DESCRIPTION: The Missile Defense Agency (MDA) seeks to leverage neuromorphic technology to maximize performance and efficiency for terrestrial and space applications. Neuromorphic processing supports multiple applications that advance the Missile Defense System, including distributed sensing and real-time learning for arbitrary waveform generation. In highly contested environments, autonomously adaptive arbitrary waveform generation supports superiority in the radio frequency domain. The communication throughput and edge processing required for distributed sensing would be realized with the low latency and high throughput inherent to neuromorphic technology.

PHASE I: Demonstrate a pathway to a product that provides solutions to the challenges of neuromorphic hardware. Phase I proposals would be evaluated to the degree to which they are applicable to the Missile Defense System, its sensing needs, and the ability to perform real-time learning. Phase I Deliverables should include:
- Assessment of the challenges to performance, packaging, and survivability
- A roadmap to develop a Technology Readiness Level (TRL) 6 product
- Identify hardware, software, and material required in Phase II
- Development, Test, and Evaluation Plan of a TRL 6 Prototype

PHASE II: Demonstrate autonomously adaptive arbitrary waveform generation using neuromorphic processing. Proposals will be evaluated based on the degree to which they demonstrate:
- Incorporation of Neuromorphic technology
- Operation in a radio frequency-contested environment
- Operation in a space environment
- High throughput
- Efficient processing
- Real-time latency

Performance categories of interest: latency, plasticity, complexity of mathematical computations, computations per second per watt, throughput, power utilization, machine speed (action/reaction). Proposals applying neuromorphic semi-conductor chips (with bio-inspired learning and parallel processing) to address these problem sets will be prioritized. Solutions could range from individual systems-on-a-chip (SoC) to complete circuit card assemblies (CCAs).

PHASE III DUAL USE APPLICATIONS: Production, test, and evaluation in a realistic environment with a system level testbed. Technology shall be ready for demonstration in relevant operational environment. This environment could include placement on airborne or space low size, weight, and power (SWaP) platforms. All technical parameters for the technology given will be verified. Autonomously adaptive arbitrary waveform generation using neuromorphic processing shall be demonstrated.

REFERENCES:

1. https://www.darpa.mil/research/programs/systems-of-neuromorphic-adaptive-plastic-scalable-electronics
2. https://afresearchlab.com/technology/nics
3. https://www.whitehouse.gov/wp-content/uploads/2025/03/Amended-National-Strategy-on-Microelectronics-Research.pdf (Page 18)
   "Progress is being made in integrating semiconductor systems with biomolecular, biological, neuromorphic, and bio-inspired systems that may one day deliver unprecedented improvements in energy efficiency and other unique capabilities in computation, AI, robotics, sensing, and health care beyond the potential of either domain on its own."- Page 18
4. https://www.cto.mil/wp-content/uploads/2025/03/TRA-Guide-Feb2025.v2-Cleared.pdf (Page 7)
   "Examples include testing a prototype in a high-fidelity laboratory environment or in a simulated operational environment" – Page 7

KEYWORDS: Neuromorphic Processor; Biology Inspired; Artificial Intelligence; Sensor; Communications; Deep Learning; Hardware; Adaptive Plastic Scalable Electronics

MDA26BZ04-NV007 TITLE: Universal Automated Control Assessment, Validation & Risk Correlation Platform for Hybrid DoW Environments

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Contested Logistics Technologies (LOG); Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Trusted AI and Autonomy; Integrated Sensing and Cyber; Human-Machine Interfaces

PROJECTED CMMC LEVEL REQUIREMENT: Level 1

OBJECTIVE: Develop a universal assessment framework capable of automating Control Validation Tests (CVT) across diverse operating environments (Cloud, On-Premise, and Air-Gapped/Tactical). The solution must ingest data from existing vulnerability scanners and automatically correlate technical findings to NIST SP 800-53 controls, reducing manual administrative overhead and allowing assessment teams to focus on mission-critical risk analysis.

DESCRIPTION: The Missile Defense Agency (MDA) requires a standardized, environment-agnostic capability to validate cybersecurity controls across its complex architecture. Current assessment methodologies rely heavily on manual data correlation—assessors spend valuable time mapping vulnerability scan results (CVEs) and STIG checklists to RMF controls (NIST 800-53). This manual process is slow, prone to inconsistency, and diverts high-value human capital from analyzing actual mission risk.
The Agency seeks an "Assessment Orchestration" solution that can:
-Operate Anywhere: Function identically in cloud-native, enterprise on-premise, and disconnected/austere environments, providing a unified data structure regardless of the target's location.
-Automate the "Grind": Ingest raw outputs from standard tools (e.g., Nessus/ACAS, SCAP) and automatically map findings to the relevant security controls (NIST 800-53, with extensibility for NIST 800-171/CMMC).
-DCO Alignment: Bridge the gap between Assessment (SCA) and Operations (DCO) by validating the implementation status of directed actions (e.g., Cyber Tasking Orders) on the target system.
-Data Portability: Ensure assessment data can be securely synchronized from tactical edge environments to strategic governance hubs for aggregation and trend analysis.

PHASE I: 1. Universal Data Ingestion: Demonstrate the feasibility of parsing and normalizing outputs from standard DoD tools (ACAS, SCAP) into a unified assessment database.
2. Automated Control Mapping: Develop algorithms to correlate technical vulnerabilities (CVEs) and configuration settings (STIGs) to specific NIST SP 800-53 controls with high accuracy.
3 Hybrid Architecture Design: Define a modular architecture that allows the core assessment engine to run effectively on a cloud instance, a local server, or a standalone laptop without code refactoring.
4. Assessor Workflow Optimization: Research and design a user experience (UX) that integrates the automated assessment results into a streamlined workflow for human validation and risk adjudication.

PHASE II: Develop, demonstrate, and pilot a functional "Assessment Orchestration" prototype based on the architecture defined in Phase I. The Phase II effort shall result in a deployable Minimum Viable Product (MVP) that demonstrates:
1. End-to-End Assessment Workflow: Demonstrate a complete, automated cyber assessment lifecycle, from initial data ingestion and automated control mapping to the final generation of valid compliance artifacts (e.g., POA&M, Security Assessment Report).
2. Longitudinal Trend Analysis: Demonstrate a centralized capability to aggregate assessment data over time, visualizing risk trends, maturity improvements, and configuration drift between assessment periods.

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3. Operational Alignment: A demonstrated interface or methodology for validating that specific Defensive Cyber Operations (DCO) mandates (e.g., CTOs, IAVMs) have been successfully applied to the target environment.

PHASE III DUAL USE APPLICATIONS: Scale the verified prototype into a mature, enterprise-ready capability for broad deployment across the MDA Enterprise and the Defense Industrial Base (DIB).
-DoD Transition: Integration into the standard Cyber Vulnerability Team (CVT) workflow to support Continuous ATO (cATO). The solution should enable a "assess once, report many" capability, feeding valid data to enterprise GRC and DCO stakeholders.
-Commercial / DIB Transition (CMMC): Adaptation of the platform to support Cybersecurity Maturity Model Certification (CMMC) compliance for the Defense Industrial Base assessments.
-Critical Infrastructure & Private Sector: Commercialization for highly regulated private sectors (Finance, Healthcare, Energy/OT) that require rigorous compliance validation (e.g., HIPAA, ARC-AMPE) in distributed or segmented network environments.

REFERENCES:

1. NIST Special Publication 800-53 Rev. 5, "Security and Privacy Controls for Information Systems and Organizations."
2. DoD CIO Memo, "Continuous Authorization to Operate (cATO) Memo," 2022.
3. NIST Special Publication 800-171 Rev. 2, "Protecting Controlled Unclassified Information in Nonfederal Systems and Organizations."

KEYWORDS: Automated Assessment; Cybersecurity; NIST 800-53; Risk Management Framework (RMF); Disconnected Operations; Control Validation; DCO Alignment; CMMC

MDA26BZ04-NV008 TITLE: Event-Based Sensing Hardware and Algorithms for Navigation

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Quantum and Battlefield Information Dominance (Q-BID)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Microelectronics

PROJECTED CMMC LEVEL REQUIREMENT: Level 1

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OBJECTIVE: Develop novel and innovative low size, weight, power, and cost (SWaP-C) event-based sensing navigation solution suitable for hostile flight and/or space environments that is not reliant on Global Positioning System (GPS) or Global navigation Satellite System (GNSS) inputs.

DESCRIPTION: The proposed initiative focuses on the development of event-based sensing technology that utilizes advanced neural network models to enhance the navigation capabilities of kill vehicles and satellites. This system aims to operate independently of GNSS yet synergistically with existing inertial navigation systems (INS) or units, providing critical inputs that improve overall navigational accuracy and responsiveness in dynamic environments. By leveraging neural networks, the technology will process event-driven data—such as changes in the vehicle's surroundings or mission-specific stimuli—allowing for real-time decision-making and adaptive navigation. Solutions should enhance the vehicle's ability to react to unforeseen circumstances but also allows for the integration of various sensor modalities, thereby enriching the situational awareness of the system. The performance benchmarks for this technology are set to align with the precision of current GPS standards, ensuring that the navigation capabilities meet or exceed the existing levels of accuracy required for advanced military operations. Development should plan on integration into current Missile Defense Agency platforms to maintain compatibility with established navigation frameworks.

PHASE I: Design and develop innovative solutions, methods, and concepts for an advanced visual navigation system utilizing event-based sensing technologies capable of low-earth orbit and/or extremely highspeed flight environments. Deliverables will include, description and data supporting initial design through modeling, experimentation, and/or testing.

PHASE II: -Refine and document detailed requirements in collaboration with the platform(s) system integrator.
-Develop a functional system prototype or algorithm able to perform on representative hardware.
-Conduct comprehensive testing of prototypes in representative operational environments.
-Demonstrate successful navigation operation without GPS inputs.

PHASE III DUAL USE APPLICATIONS: -Define clear navigation system requirements and interface specifications in collaboration with the system integrator.
-Produce flight ready prototype in accordance with test required volumes, power needs and interfaces.

-Demonstrate successful navigation outputs during powered flight that mirror GPS within acceptable tolerances stated by platform engineers and security allowances.

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KEYWORDS: Event-Based; Visual; Navigation; SIFT; FLANN Matcher; YOLO; Neural Network; VIO systems; GPS; INS; Inertial; Keypoints; Landmarks; Star tracker; Celestial